



香港中文大學(深圳)

The Chinese University of Hong Kong, Shenzhen

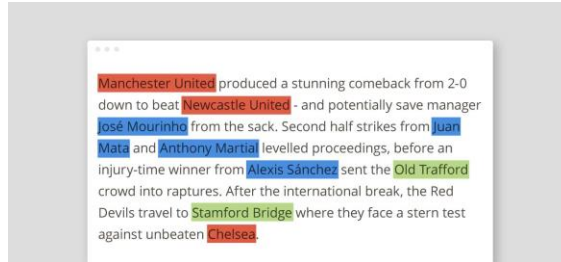
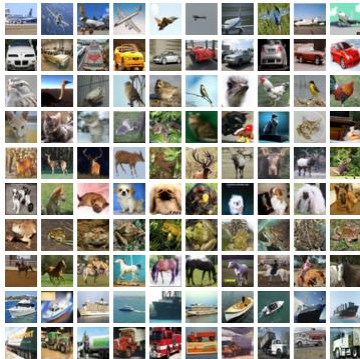
Introduction to Data Science

Lecture 21 Machine Learning: Introduction & KNN

What is Machine Learning?

Machine Learning (ML) is a task-driven field of computer science that focuses on building systems that **learn from data** to improve performance on a specific **task** through **experience** — without being explicitly programmed.

Diverse forms of input data including numbers, words, clicks and images.



Core Formula (from Tom Mitchell):

A computer program is said to learn from **experience** E , with respect to some **task** T , and some **performance measure** P , if its performance on T , as measured by P , improves with experience E .

Key Ingredients:

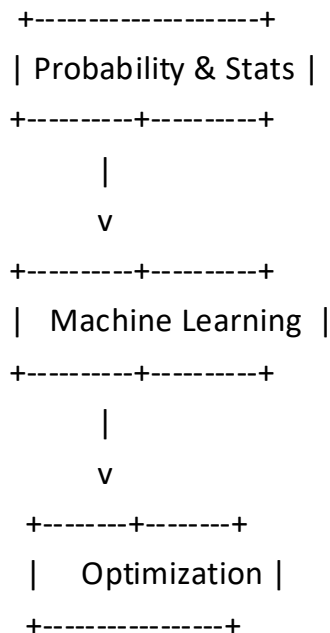
- **Task:** What do we want the machine to do? (e.g., classify, predict)
- **Data:** Experience (input-output pairs or raw observations)
- **Model:** A function that maps input to output
- **Learning:** Improve the model using data (optimize performance)

*ML systems **generalize** from data to solve new instances of a task.*

How ML Connects with What We've Learned

Concept	Role in ML
Probability	Modeling uncertainty (e.g., probabilistic models, Bayes' theorem)
Statistics	Understanding and analyzing data; model evaluation (e.g., bias/variance)
Optimization	Training models by minimizing loss

Conceptual Diagram:



Takeaway:

ML stands at the intersection of these fields — it uses data (statistics), models uncertainty (probability), and trains models (optimization).

Resources

Browse > Data Science > Machine Learning

Offered By

Machine Learning

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 Andrew Ng **TOP INSTRUCTOR**

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Stanford

<https://github.com/kjaisingh/high-school-guide-to-machine-learning>

What Determines the Performance of a Machine Learning Model?

The performance of a machine learning model largely depends on two key factors:

1. Quality of Input Data

High-quality, well-organized data is essential. If the input data is noisy, incomplete, or inconsistent, the model is likely to produce unreliable or inaccurate predictions.

2. Choice of Model/Algorithm

Different algorithms are suited for different types of tasks and data structures. Selecting an algorithm that aligns with the problem characteristics is critical for achieving good performance.



How Does Machine Learning Work?

Step 1: Prepare the Training Data

- Gather and preprocess a representative dataset.
- This data serves as the foundation for the model to learn from.

Step 2: Select and Apply an Algorithm

- Choose an appropriate machine learning algorithm based on the problem type (e.g., classification, regression, clustering).
- Apply the algorithm to the training data.

Step 3: Train the Model

- Adjust model parameters to minimize prediction error.
- The model learns patterns in the data automatically, with minimal human intervention — this is the core of machine learning.

Step 4: Deployment and Model Improvement

- Deploy the trained model to make real-world predictions.
- Continuously improve the model with new data and periodic retraining to maintain or enhance accuracy.

Next, we will discuss the first machine learning task that will be discussed in this course– **classification**

Classification

What is a Classification Task?

- **Goal:** Assign an input to one of a set of **discrete categories (classes)**.

Examples:

- Email → Spam or Not Spam
- Image → Cat, Dog, or Bird
- Patient Data → Disease A, Disease B, or Healthy

Input: Feature vector (e.g., [height, weight, age])

Output: Class label (e.g., "overweight" or "healthy")

Example

Is it a cat or a Dog?



{ cat
dog

Example

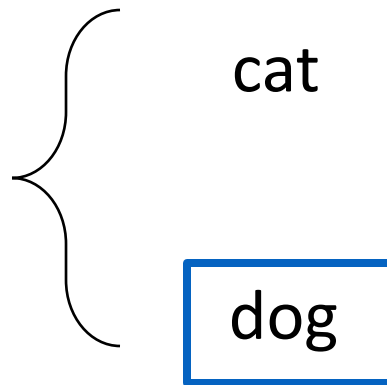
Is There a Cat or a Dog?



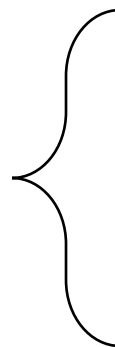
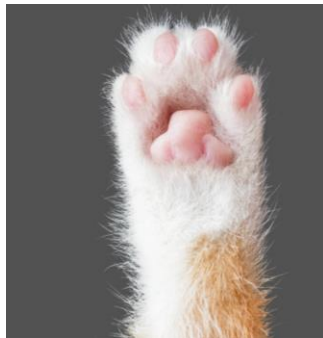
{ cat
dog

Example

Is it a cat or a Dog?



How about more challenging cases?



cat

?

dog

Represent Objects Numerically

From Pictures to Numbers (Feature Representation)

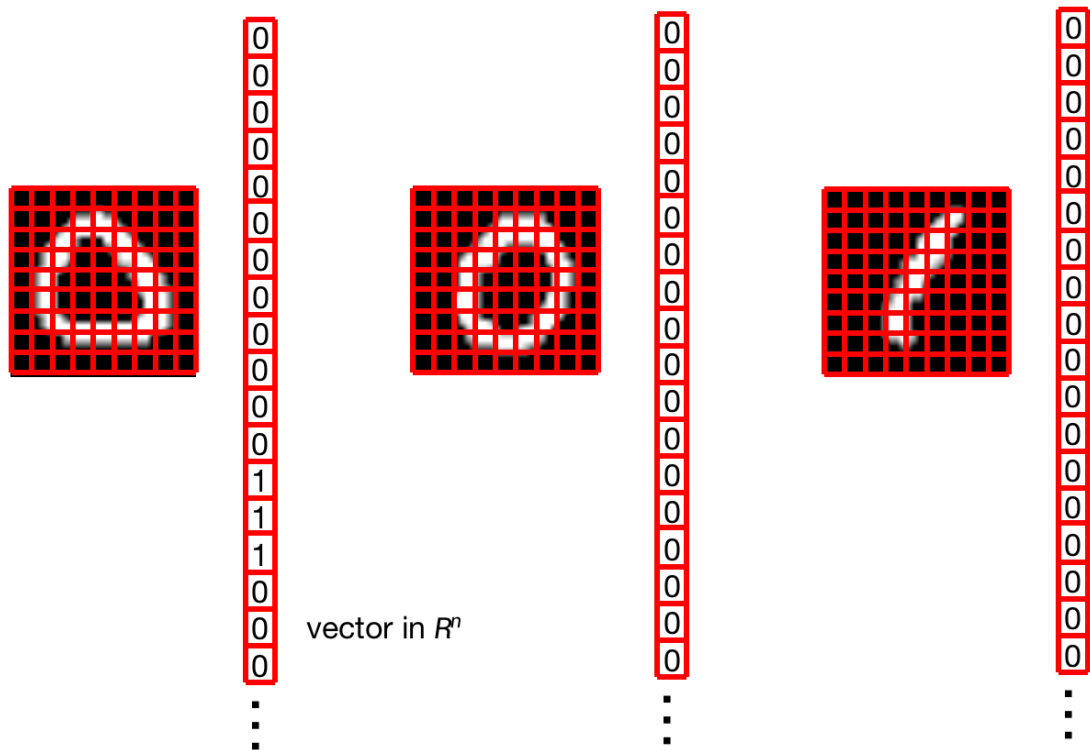
Problem: Can we teach a computer to tell cats and dogs apart in pictures?

To a human, it's easy:

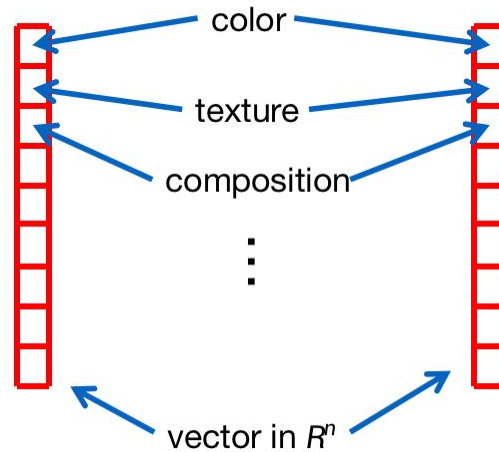
-  Pointy ears, small nose
-  Floppy ears, big snout

But to a computer, we must turn the image into **numbers**!

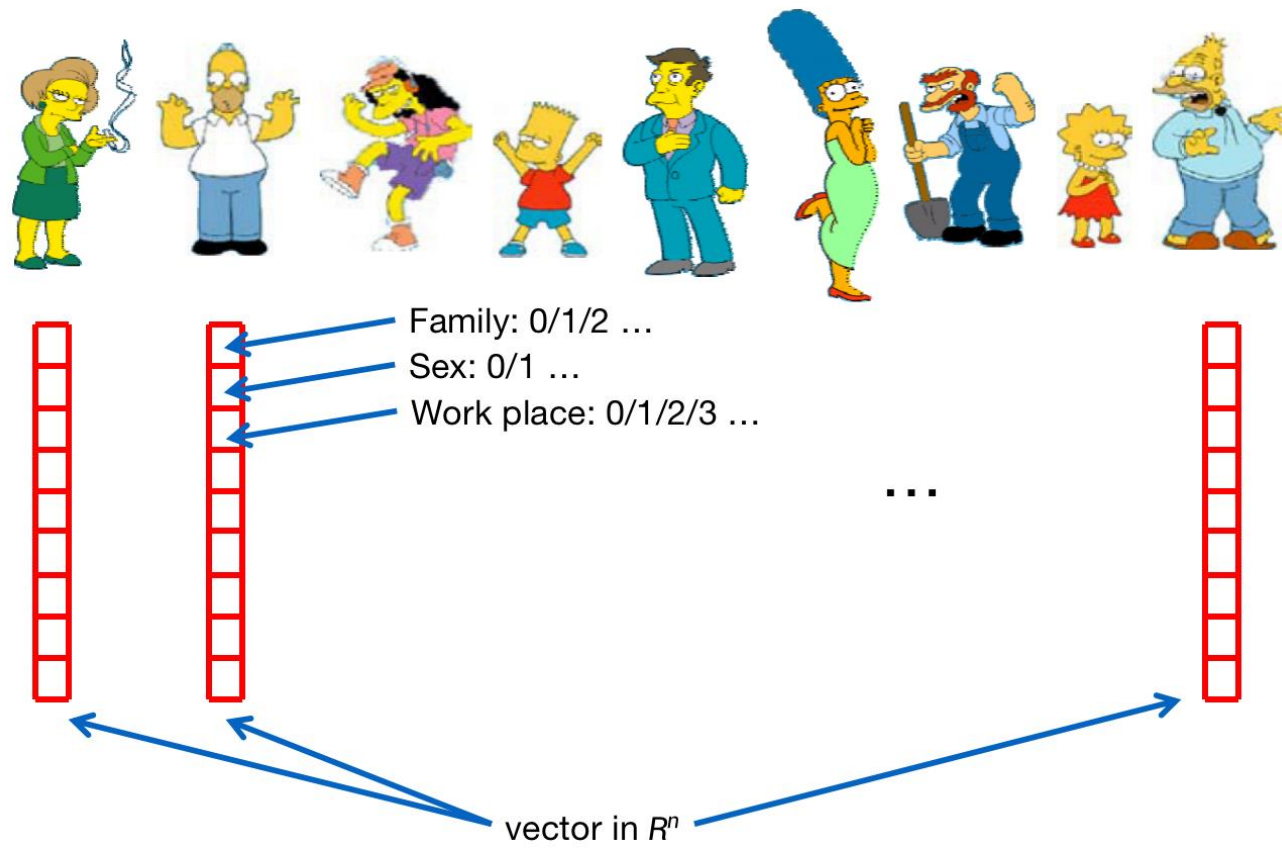
How to represent objects numerically?



Images of different sizes



Objects in real life



Supervised Learning vs. Unsupervised Learning

Supervised vs. Unsupervised Learning (with Cats & Dogs)

Supervised Learning – we show examples *with answers*



"Here's a cat" → model learns



"Here's a dog" → model learns

Then it predicts new ones!

Unsupervised Learning – no labels, just data



"Here are a bunch of animals"

Model tries to group similar ones (maybe cats in one group, dogs in another)

Supervised vs. Unsupervised Learning (with Cats & Dogs)

	Supervised	Unsupervised
We provide:	Cat/Dog labels 🏷️	No labels 🐒
Model goal:	Learn to predict 🧠	Find patterns 🔍
Cat & Dog Example:	"This is a cat" 🐱	"These look similar..." 🐾🐾

Supervised learning is a type of machine learning where the algorithm is trained on a **labeled** dataset.

Unsupervised learning is a type of machine learning where the algorithm is given input data **without any labels**.

Supervised Learning for Classification

- **Supervised machine learning** algorithms utilize **labeled data** for training, where the correct outputs corresponding to input data are already known.
- For all samples, (x^i, y^i) , $i = 1, \dots, N$, you can observe both the input data x^i and the label y^i

Training data



$y=1$ (cat)



$y=0$ (dog)



$y=1$ (cat)

...



$y=0$ (dog)

Supervised Learning for Classification

Training data (with labels)



$y=1$ (cat)



$y=0$ (dog)

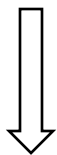


$y=1$ (cat)

...



$y=0$ (dog)



(Build a classifier)

Classifier $h: X \rightarrow \{0,1\}$

Given a new data, predict the label



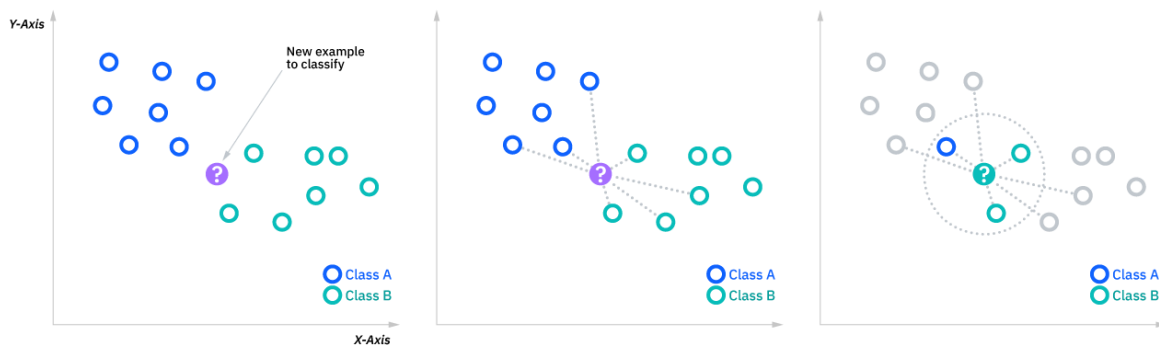
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Meet Your First ML Algorithm — A Supervised Learning Algorithm for Classification Task:

k-Nearest Neighbors (KNN)

K-Nearest Neighbor Classifier

- Find K training points x_i closest to x .
- If the majority of K -nearest neighbors of x belong to classifier c , label x as c .



The KNN Algorithm

1. Load the data
2. Set K of your choice to be the number of neighbors
3. For each new data to be classified
 - Calculate the distances between the new data and all the labeled data.
 - Record the entry (d_i, y_i) , where d_i is the distance between the new data and the i -th labeled data, and y_i is the label of the i -th data.
 - Sort the these entries with respect to distance (from smallest to largest).
4. Pick the first K entries from the sorted collection
5. Get the labels of the selected K entries
6. Choose the label with the largest frequency

Example 1

甜豆腐脑派 VS 咸豆腐脑派



Sweet tofu pudding or salted tofu pudding?

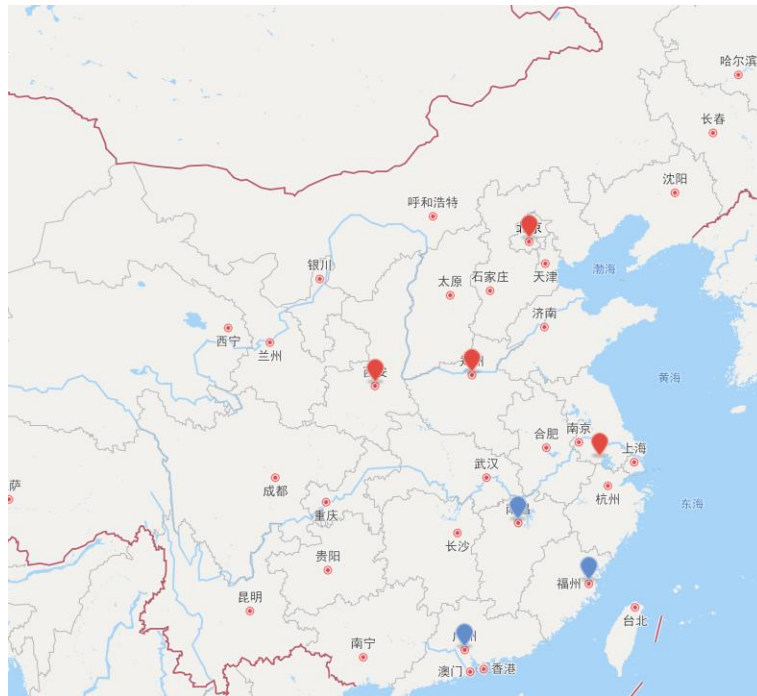
Example 1

Data:

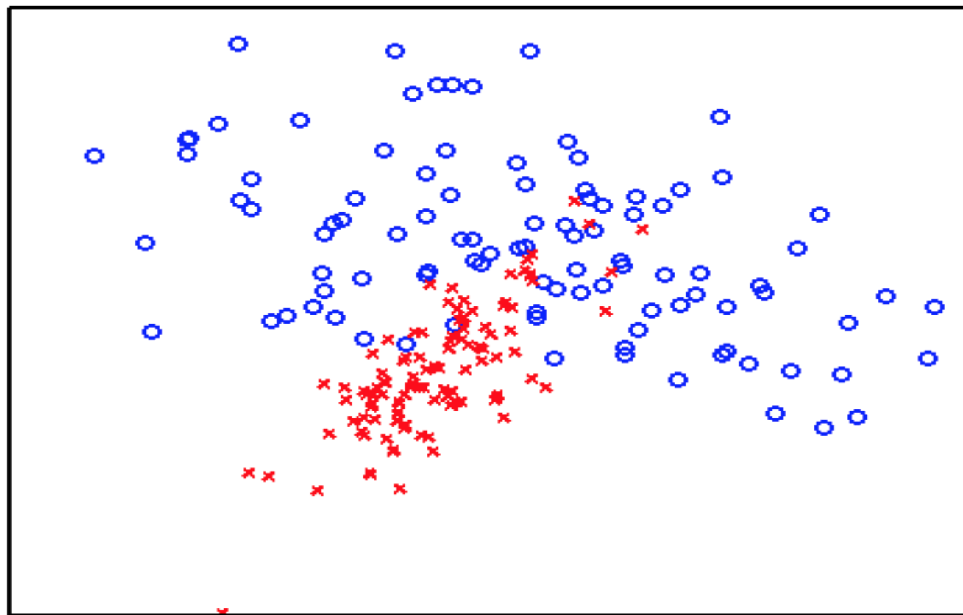
Your hometown location

Label:

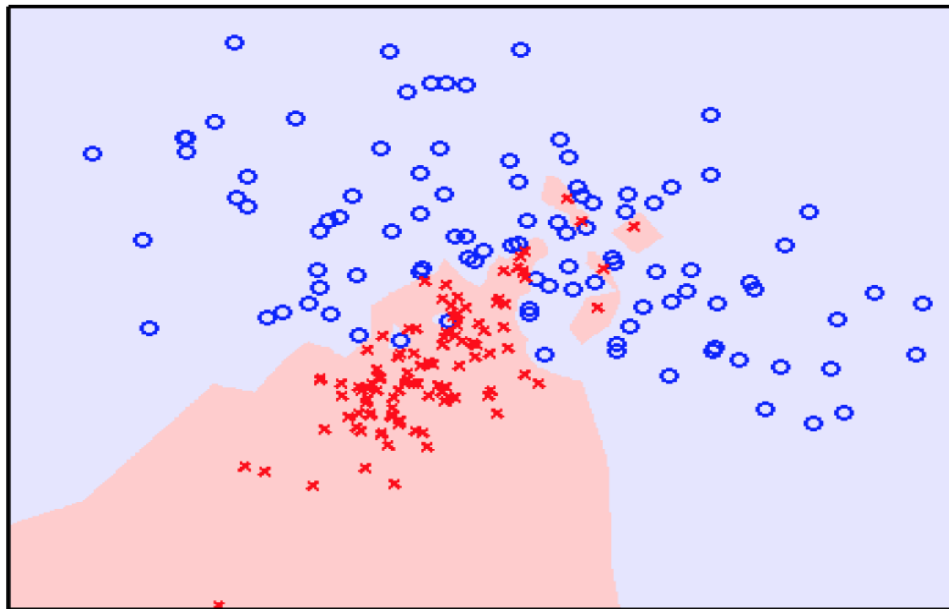
Red (salted)/Blue (Sweet)



Example 2

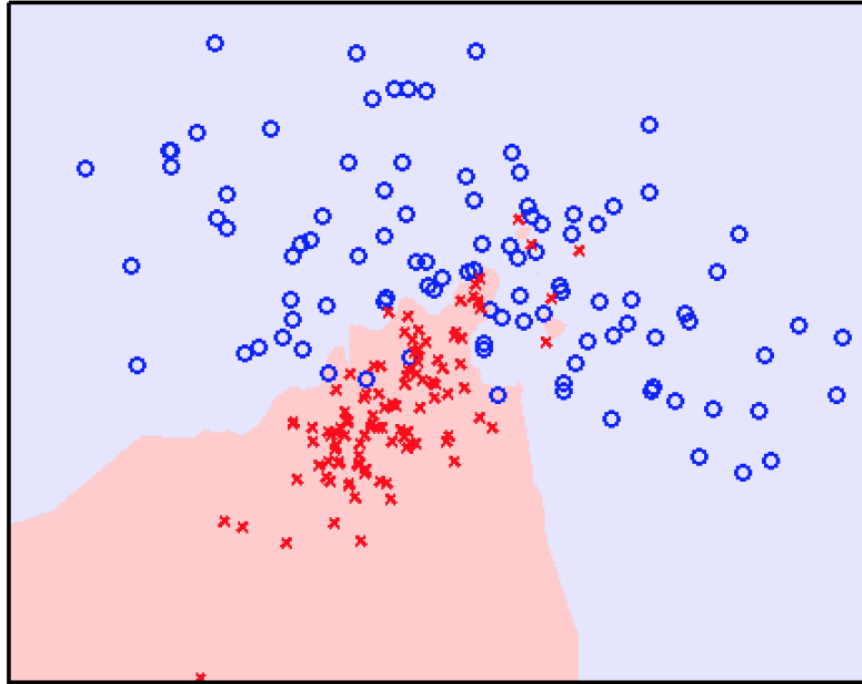


Example 2



$$K = 1$$

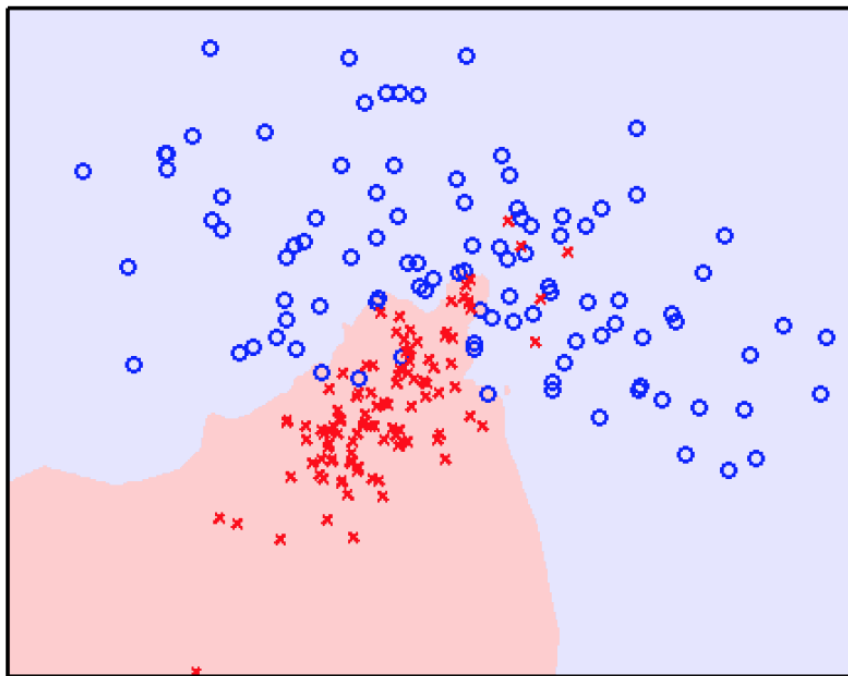
Example 2



$K = 3$

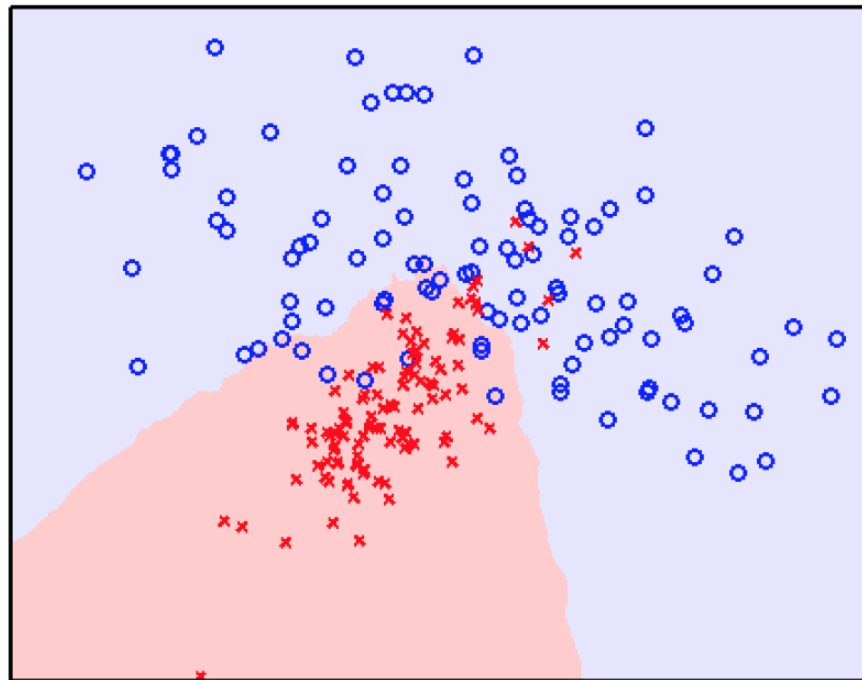
The red/blue region indicates the area where any new data falling within will be classified as 'red/blue' by the algorithm.

Example 2



$K = 5$

Example 2



$K = 25$

- The selection of K is critical—lower values of K may significantly increase the influence of noise on the outcomes.
- A large value of K may undermine the fundamental principles underlying the K -Nearest Neighbors algorithm.
 - If $K > \text{total number of labeled data}$, the label of any new input will be the labels that appears the most in the samples.

Exercise

- We are given the following data set with points of three different classes:

Points	x_1	x_2	class
A	0	0	1
B	-3	1	1
C	5	2	2
D	3	3	2
E	5	0	3
F	4	-1	3

We perform a K -NN classification. Classify the new point $(4, 3)$ with $K = 1$ using the L_1 -norm as the distance measure.

L_1 norm distance/Manhattan distance between x and y : $|x_1 - y_1| + |x_2 - y_2|$

Points	x_1	x_2	class
A	0	0	1
B	-3	1	1
C	5	2	2
D	3	3	2
E	5	0	3
F	4	-1	3

We perform a K -NN classification. Classify the new point $(4, 3)$ with $K = 1$ using the L_1 -norm as the distance measure.

$$d(A, \text{new point}) = |0 - 4| + |0 - 3| = 7$$

$$d(B, \text{new point}) = |-3 - 4| + |1 - 3| = 9$$

$$d(C, \text{new point}) = |5 - 4| + |2 - 3| = 2$$

$$d(D, \text{new point}) = |3 - 4| + |3 - 3| = 1$$

$$d(E, \text{new point}) = |5 - 4| + |0 - 3| = 4$$

$$d(F, \text{new point}) = |4 - 4| + |-1 - 3| = 4$$